

Fundamentals of Linear Control:

A Concise Approach

Maurício C. de Oliveira

Chapter 7: Frequency Domain

linearcontrol.info/fundamentals

7.1 Bode Plots

Goal

Easily sketch the frequency response of rational transfer-function models

Setup

Given
$$G(s) = \frac{\beta \cdot (s - z_1)^{\ell_1} (s - z_2)^{\ell_2} \cdots (s - z_r)^{\ell_r}}{s^{k_0} (s - p_1)^{k_1} (s - p_2)^{k_2} \cdots (s - p_t)^{k_t}}$$
 rewrite
$$G(s) = \frac{\tilde{\beta} \cdot (1 + \sigma_1 s)^{\ell_1} (1 + \sigma_2 s)^{\ell_2} \cdots (1 + \sigma_r s)^{\ell_r}}{s^{k_0} (1 + \tau_1 s)^{k_1} (1 + \tau_2 s)^{k_2} \cdots (1 + \tau_t s)^{k_t}}$$
 where $\sigma_j = -z_j^{-1}$, $\tau_i = -p_i^{-1}$, $\tilde{\beta} = (-1)^{m-n} \frac{\beta \cdot z_1^{\ell_1} \cdots z_r^{\ell_r}}{p_1^{k_1} \cdots p_t^{k_t}}$

Setup

Normalized transfer-function

$$\begin{aligned} G(s) = \frac{\tilde{\beta}}{(1 + \sigma_1 s)^{\ell_1} (1 + \sigma_2 s)^{\ell_2} \cdots (1 + \sigma_r s)^{\ell_r}} s^{k_0} (1 + \tau_1 s)^{k_1} (1 + \tau_2 s)^{k_2} \cdots (1 + \tau_t s)^{k_t} \end{aligned}$$

Magnitude

$$\begin{aligned} 20 \log_{10} |G(j\omega)| &= 20 \log_{10} |\tilde{\beta}| - 20 \log_{10} |j\omega|^{k_0} \\ &\quad + 20 \log_{10} |1 + j\sigma_1 \omega| + \cdots + 20 \log_{10} |1 + j\sigma_r \omega| \\ &\quad - 20 \log_{10} |1 + j\tau_1 \omega| - \cdots - 20 \log_{10} |1 + j\tau_t \omega| \end{aligned}$$

Phase

$$\begin{aligned} \angle G(j\omega) &= \angle \tilde{\beta} - k_0 \angle j\omega \\ &\quad + \ell_1 \angle (j\sigma_1 \omega + 1) + \cdots + \ell_r \angle (j\sigma_r \omega + 1) \\ &\quad - k_1 \angle (j\tau_1 \omega + 1) - \cdots - k_t \angle (j\tau_t \omega + 1) \end{aligned}$$

First-order real poles and zeros

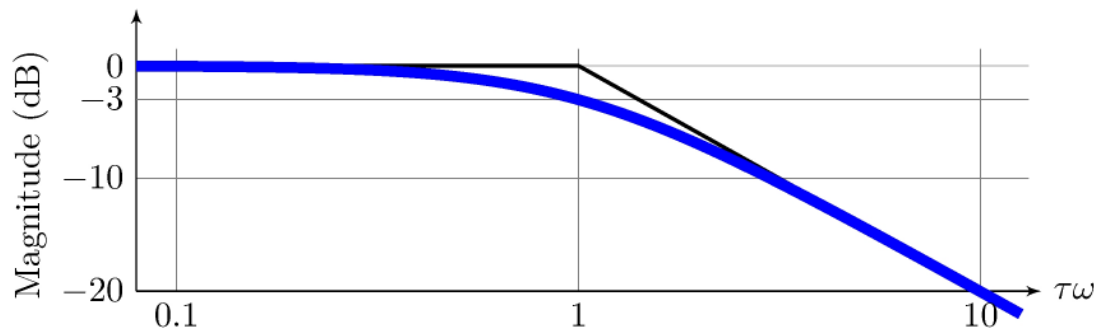
Transfer-function

$$G(s) = \frac{1}{(1 + s\tau)^k}$$

Magnitude

$$20 \log_{10} |G(j\omega)| \approx \begin{cases} 0, & |\omega| < \tau^{-1} \\ -20k \log_{10} |\omega|, & |\omega| \geq \tau^{-1} \end{cases}$$

Example: simple pole ($\tau = 1$, $k = 1$)



First-order real poles and zeros

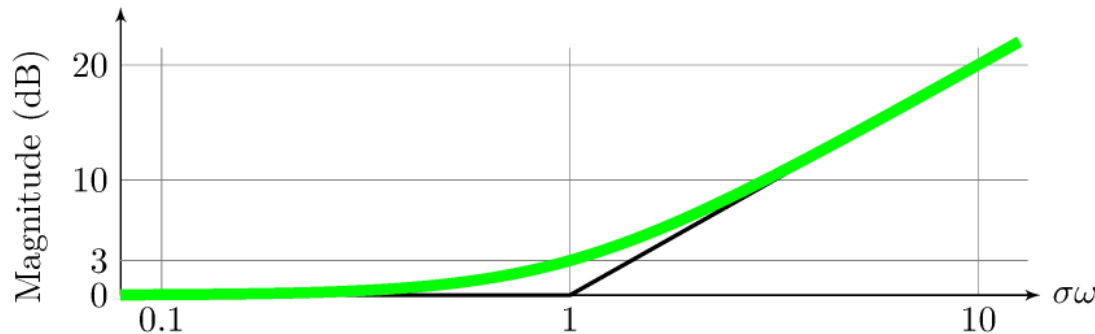
Transfer-function

$$G(s) = (1 + s\tau)^k$$

Magnitude

$$20 \log_{10} |G(j\omega)| \approx \begin{cases} 0, & |\omega| < \tau^{-1} \\ 20k \log_{10} |\tau| + 20k \log_{10} |\omega|, & |\omega| \geq \tau^{-1} \end{cases}$$

Example: simple zero ($\tau = \pm 1$), ($k = 1$)



First-order real poles and zeros

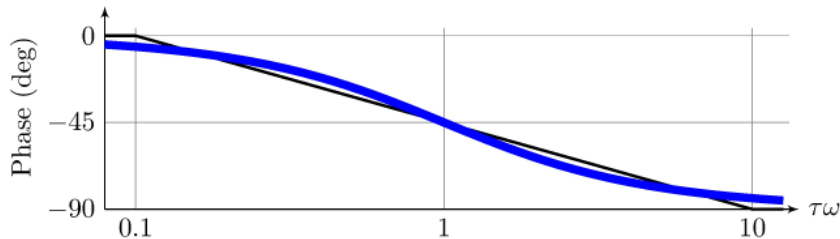
Transfer-function

$$G(s) = \frac{1}{(1 + s\tau)^k}$$

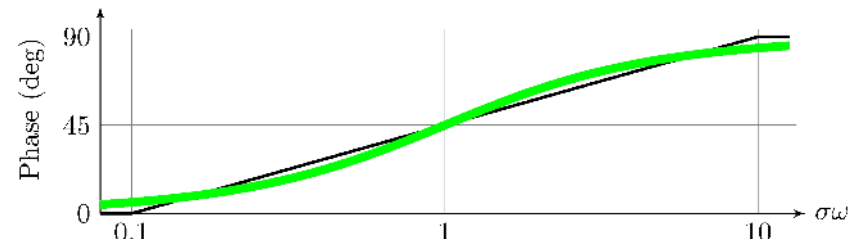
Phase

$$\angle G(j\omega) = -k \angle (1 + j\omega\tau) \approx \begin{cases} -\frac{k\pi}{4} \log_{10} \frac{\omega\tau}{1} & 0 < \omega\tau < 10 \\ -\frac{k\pi}{2} & \omega\tau \geq 10 \end{cases}$$

Example: simple pole ($\tau = 1$), ($k = 1$)



Example: simple zero ($\tau = -1$), ($k = 1$)



First-order real poles and zeros

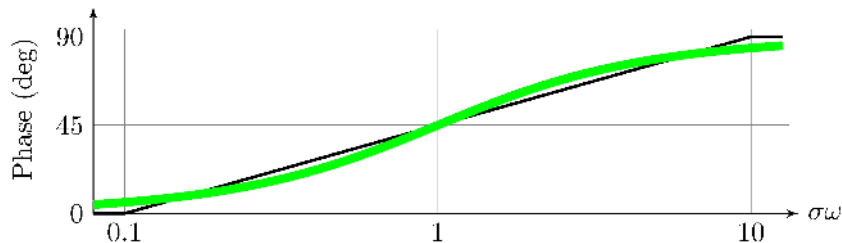
Transfer-function

$$G(s) = (1 + s\tau)^k$$

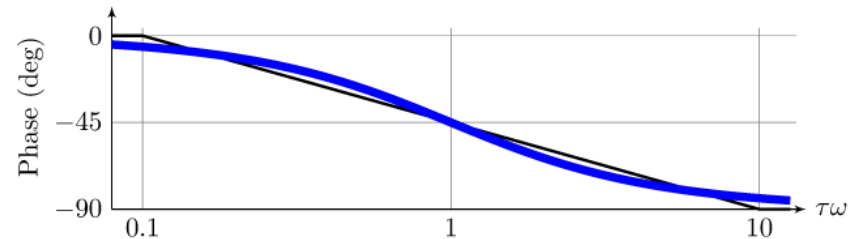
Phase

$$\angle G(j\omega) = k \angle (1 + j\omega\tau) \approx \begin{cases} 0, & |\omega| < \frac{1}{10|\tau|^{-1}} \\ \frac{k\pi}{4} \log_{10} \frac{|\tau|\omega}{1}, & \frac{1}{10|\tau|^{-1}} \leq \omega < 10|\tau|^{-1} \\ \frac{k\pi}{2}, & \omega \geq 10|\tau|^{-1} \end{cases}$$

Example: simple zero ($\tau = 1$), ($k = 1$)



Example: simple pole ($\tau = -1$), ($k = 1$)



Second-order complex poles and zeros

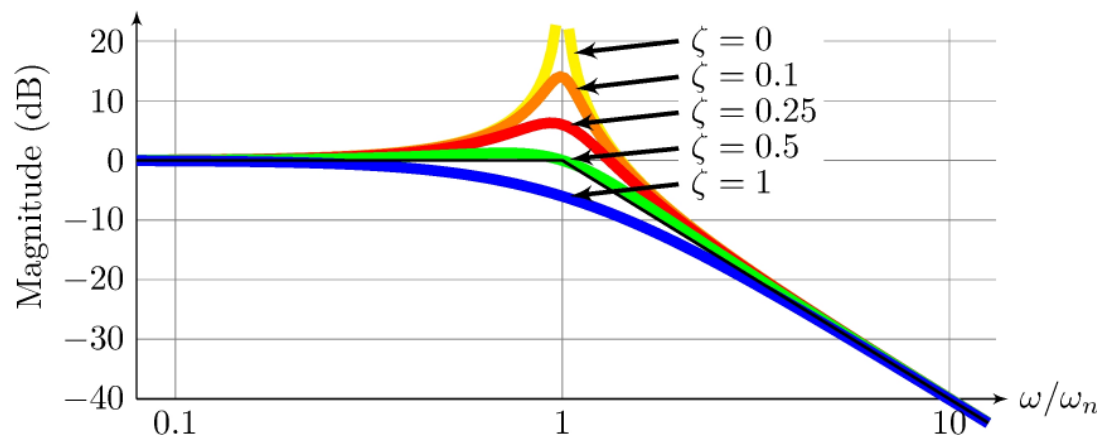
Transfer-function

$$G(s) = \frac{1}{(1 + 2\zeta s/\omega_n + s^2/\omega_n^2)^k}, \quad \omega_n > 0, \quad |\zeta| < 1$$

Magnitude

$$20 \log_{10} |G(j\omega)| = -20k \log_{10} |1 - \omega^2/\omega_n^2 + j2\zeta\omega/\omega_n| \approx \begin{cases} 0, & |\omega| < \omega_n \\ -40k \log_{10} |\omega|, & |\omega| \geq \omega_n \end{cases}$$

Example: complex pair of poles $(\omega_n = 1, k = 1)$



Second-order complex poles and zeros

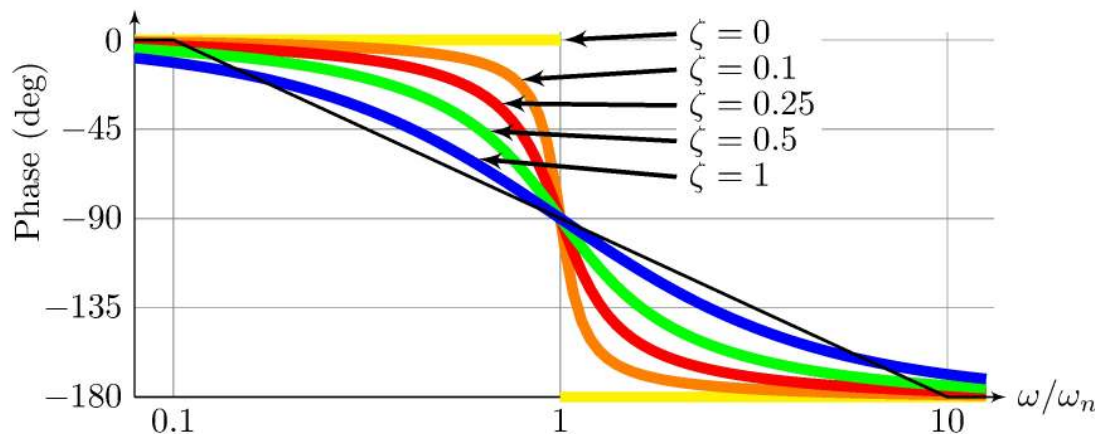
Transfer-function

$$G(s) = \frac{1}{(1 + 2\zeta s/\omega_n + s^2/\omega_n^2)^k}, \quad \omega_n > 0, \quad |\zeta| < 1$$

Phase

$$\angle G(j\omega) \approx \begin{cases} 0, & |\omega| < \frac{1}{10}\omega_n \\ -k\pi, & \omega_n \leq \omega < 10\omega_n \\ -k\pi, & \omega \geq 10\omega_n \end{cases}$$

Example: complex pair of poles $(\omega_n = 1, k = 1)$



Poles and zeros at the origin

Transfer-function

$$G(s) = \frac{1}{s^k}$$

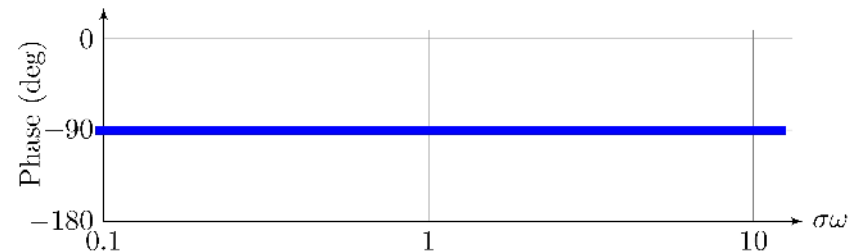
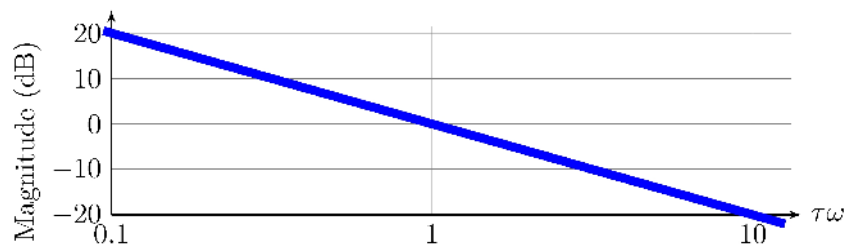
Magnitude

$$20 \log_{10} |G(j\omega)| = -20k \log_{10} |\omega|$$

Phase

$$\angle G(j\omega) = -k \frac{\pi}{2}$$

Example: simple pole ($k = 1$)



Example: (C_{lead}) from Section 6.5

Transfer-function

$$G(s) = C_{\mathrm{lead}}(s) = 3.83 \frac{s + 7.5}{s + 21}$$

Normalized

$$G(s) = \tilde{\beta} \frac{1 + s/7.5}{1 + s/21}, \quad \tilde{\beta} = \frac{3.83 \times 7.5}{21} \approx 1.37$$

Magnitude slopes

$$0 \text{ dB/decade}, \quad 0 \leq \omega \leq 7.5 \quad +20 \text{ dB/decade}, \quad 7.5 \leq \omega \leq 21 \quad 0 \text{ dB/decade}, \quad \omega \geq 21$$

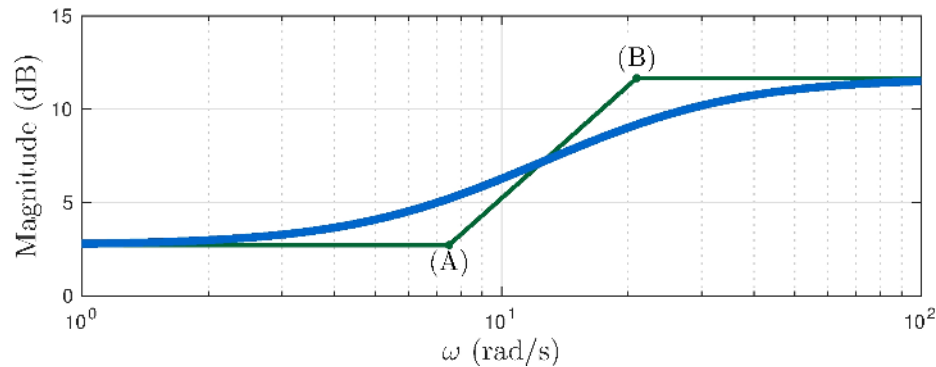
Magnitude key points

$$\begin{aligned} \tag{A} 20 \log_{10} 1.37 &\approx 2.7 \text{ dB}, & \omega \leq 7.5 & \tag{B} 2.7 \text{ dB} + 20 \log_{10} 21/7.5 \\ &\approx 11.6 \text{ dB}, & \omega \geq 21 \end{aligned}$$

Example: (C_{lead}) from Section 6.5

Transfer-function

$$G(s) = C_{\text{lead}}(s) = 3.83 \frac{s + 7.5}{s + 21}$$



Magnitude slopes

$$\begin{aligned} &0 \text{ dB/decade}, & 0 \leq \omega &\leq 7.5 \\ &+20 \text{ dB/decade}, & 7.5 \leq \omega &\leq 21 \\ &0 \text{ dB/decade}, & \omega \geq 21 \end{aligned}$$

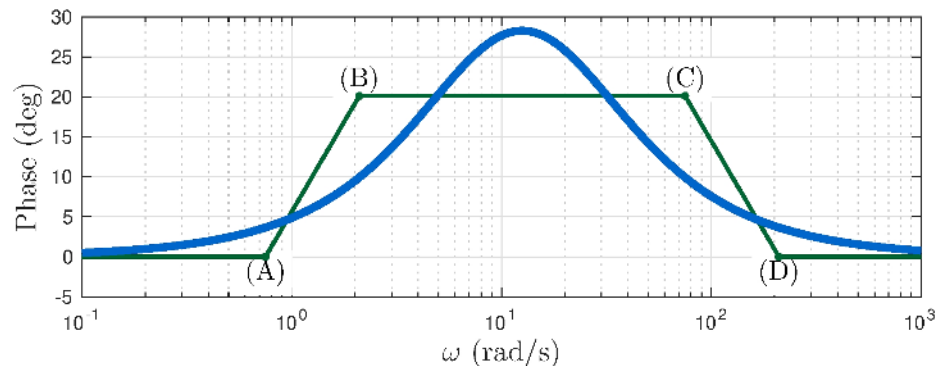
Magnitude key points

$$\begin{aligned} &\text{tag{A}} \quad 20 \log_{10} 1.37 \approx 2.7 \text{ dB}, & \omega \leq 7.5 \\ &\text{tag{B}} \quad 2.7 \text{ dB} + 20 \log_{10} 21/7.5 \approx 11.6 \text{ dB}, & \omega \geq 21 \end{aligned}$$

Example: (C_{lead}) from Section 6.5

Transfer-function

$$G(s) = C_{\text{lead}}(s) = 1.37 \frac{1 + s/7.5}{1 + s/21}$$



Phase slopes

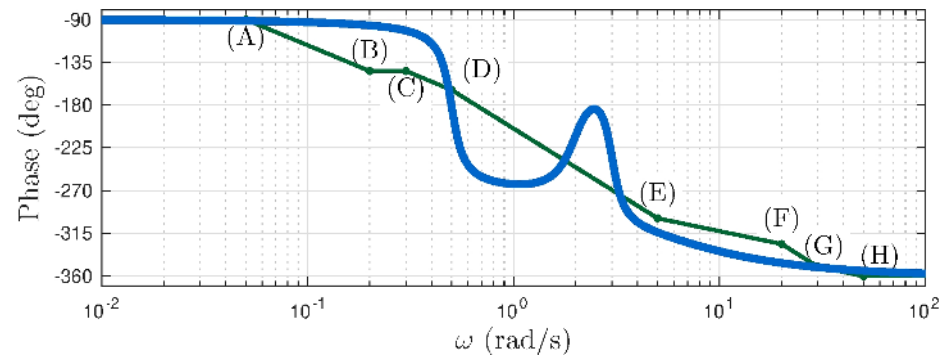
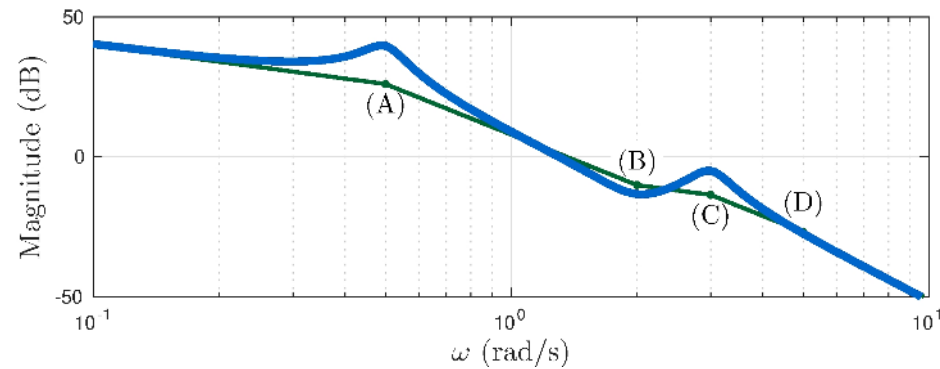
$$\begin{aligned} & 0^\circ/\text{decade}, & 0 \leq \omega & \leq 0.75 \\ & +45^\circ/\text{decade}, & 0.75 \leq \omega & \leq 2.1 \\ & 0^\circ/\text{decade}, & 2.1 \leq \omega & \leq 75 \\ & -45^\circ/\text{decade}, & 75 \leq \omega & \leq 210 \\ & 0^\circ/\text{decade}, & \omega \geq 210 \end{aligned}$$

Phase key points

$$\begin{aligned} & \text{tag{A}} \approx 0^\circ, & \omega & = 0.75 \\ & \text{tag{B}} \approx 20^\circ, & \omega & = 2.1 \\ & \text{tag{C}} \approx 20^\circ, & \omega & = 75 \\ & \text{tag{D}} \approx 0^\circ, & \omega & = 210. \end{aligned}$$

See book for a more challenging example

$$G(s) = \frac{28.1 s^2 + 22.4 s + 112.4}{s^6 + 5.7 s^5 + 12.81 s^4 + 47.6 s^3 + 7.5 s^2 + 11.25 s}$$



7.2 Non-Minimum-Phase Systems

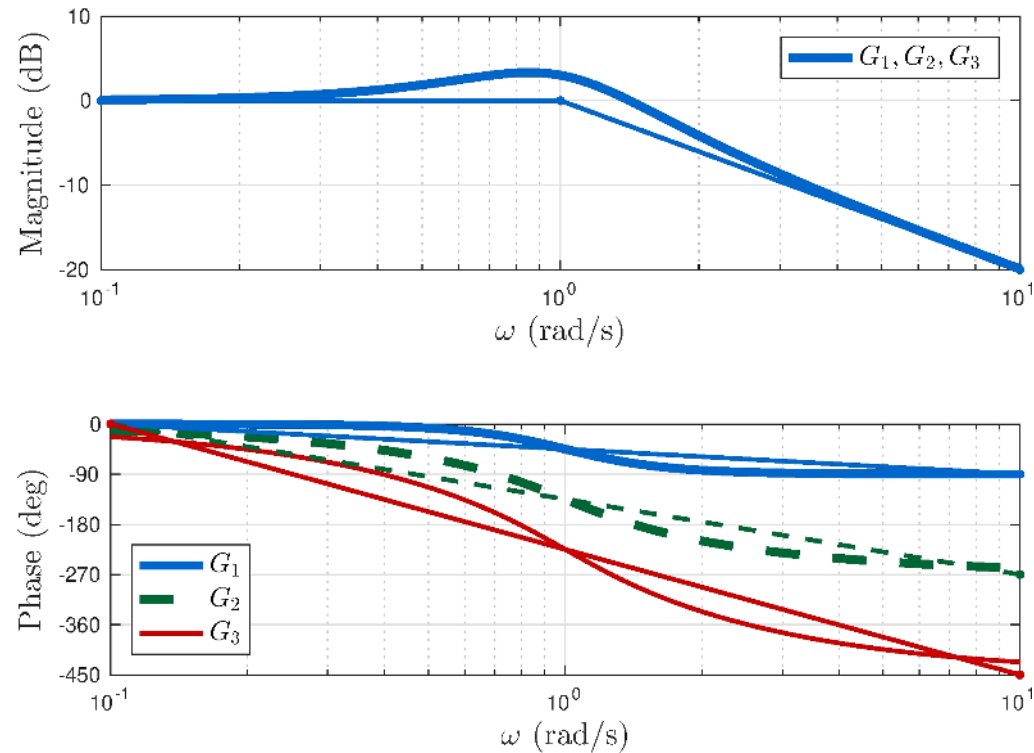
A simple example

Frequency responses of $G_1 = 1$, $G_2 = -1$, $G_3 = \frac{s-1}{s+1}$ have the exact same magnitude $|G_1(j\omega)| = |G_2(j\omega)| = |G_3(j\omega)| = 1$ but different phases $\angle G_1(j\omega) = 0$, $\angle G_2(j\omega) = \pi$, $\angle G_3(j\omega) = \pi - 2 \tan^{-1} \omega$. (G_1) is **minimum phase**

A more complex example

Frequency response of $G_1 = \frac{1+s}{s^2+s+1}$, $G_2 = \frac{1-s}{s^2+s+1}$, $G_3 = \frac{(1-s)^2}{s^3+2s^2+2s+1}$

Bode plots

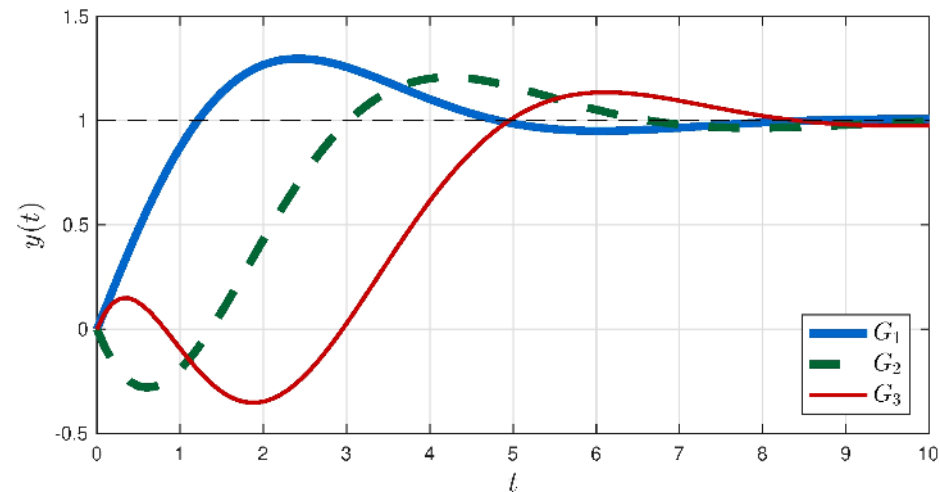


A more complex example

Frequency response of $G_1 = \frac{1+s}{s^2+s+1}$, $G_2 = \frac{1-s}{s^2+s+1}$, $G_3 = \frac{(1-s)^2}{s^3+2s^2+2s+1}$ have the exact same magnitude $|G_1(j\omega)| = |G_2(j\omega)| = |G_3(j\omega)|$

G_1 is minimum phase

Step response



Other examples

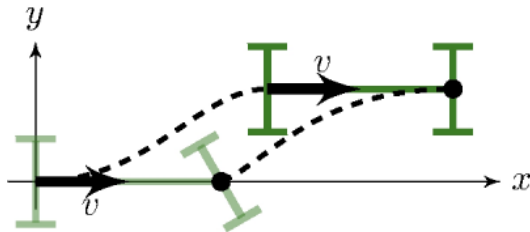
Systems with delays

Padé approximation of order (n, n)

$$e^{-s\tau} \approx \frac{1 - a_1 s + a_2 s^2 - \dots \pm a_n s^n}{1 + a_1 s + a_2 s^2 + \dots + a_n s^n}$$

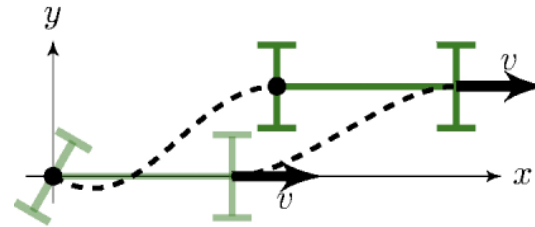
Steering vehicles

Forward lane-change



Minimum-phase

Backward lane-change



Non-minimum-phase

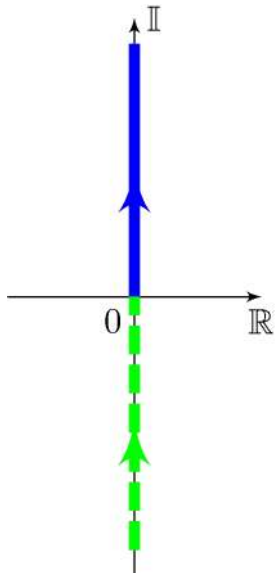
7.3 Polar Plots

Parametric curve

$$s = G(j\omega), \quad \omega \in \mathbb{R}$$

Image of the imaginary axis under the mapping G

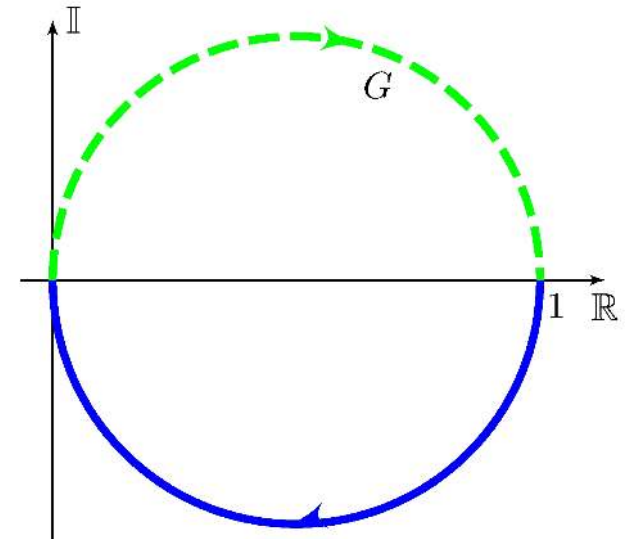
Imaginary axis



Mapping

$$G(s) = \frac{1}{s + 1}$$

Image



From Bode plots to polar plots

Example

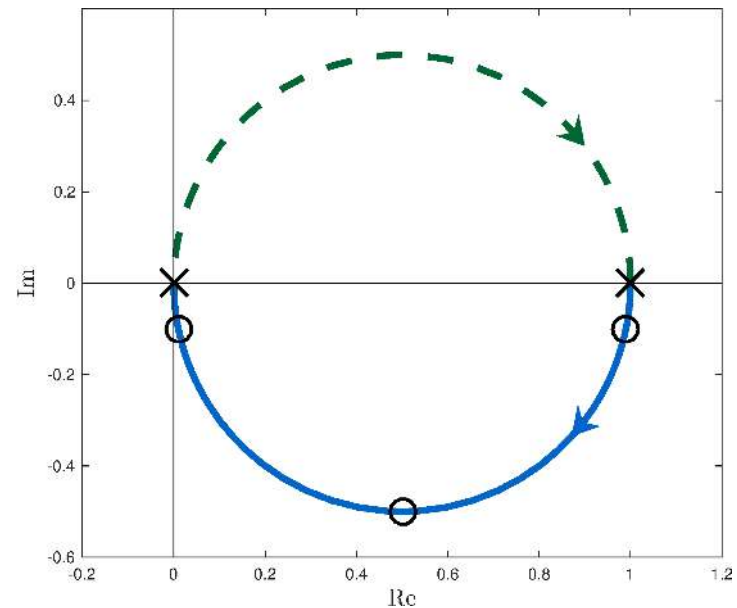
$$G(s) = (s + 1)^{-1}$$

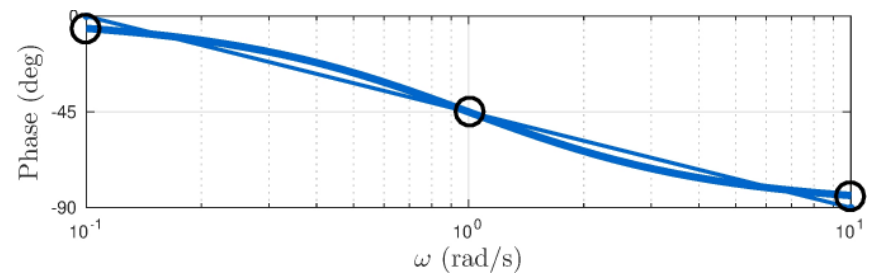
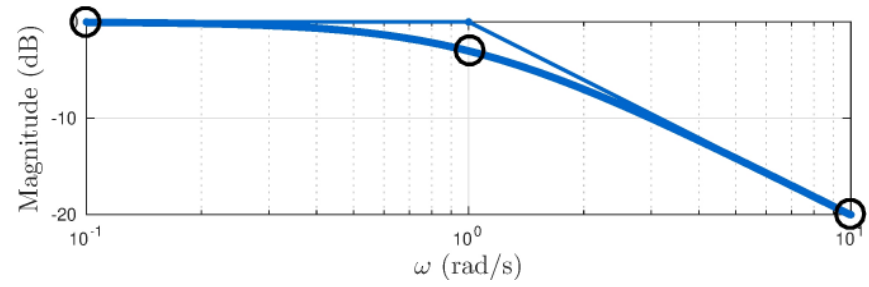
Polar plot

$$G(j\omega) = \frac{1}{j\omega + 1} = \frac{1}{2} + \frac{1}{2} \frac{1 - j\omega}{1 + j\omega} = \frac{1}{2} + \frac{1}{2} e^{j\theta}, \quad -\pi < \theta = -2 \tan^{-1} \omega < \pi$$

Bode plots

Polar plot



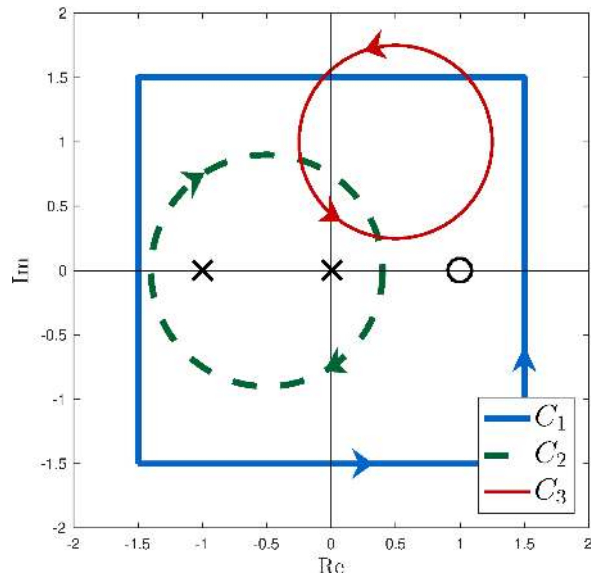


7.4 The Argument Principle

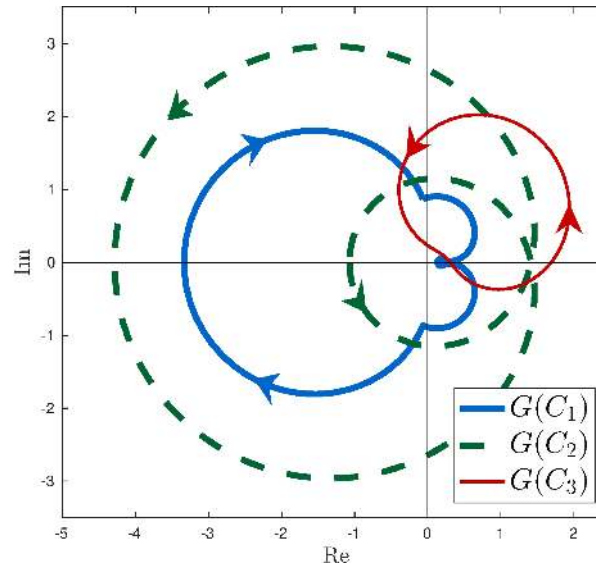
Total argument variation

$\Delta_C \arg G(s)$: total argument variation of the image of C under G about the origin

Contours



Images



Total argument

$$\begin{aligned} \frac{1}{2\pi} \Delta_{C_1} \arg G(s) &= -1 \\ \frac{1}{2\pi} \Delta_{C_2} \arg G(s) &= 2 \\ \frac{1}{2\pi} \Delta_{C_3} \arg G(s) &= 0 \end{aligned}$$

The Argument Principle

Theorem 7.1

$$\frac{1}{2\pi} \Delta_C^0 \arg f(s) = \frac{1}{2\pi} \int_C \frac{f'(s)}{f(s)} ds = Z_C - P_C$$

Z_C is the number of zeros inside C

P_C is the number of poles inside C

Assumptions

C is a positively oriented simple closed contour

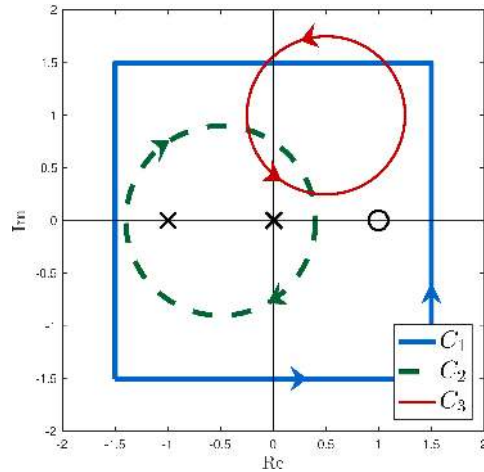
f has no zeros on C

f is analytic inside and on C except at a finite number of poles inside C

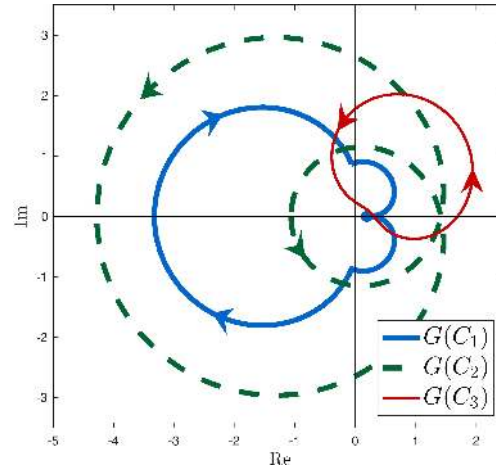
If contour is negatively oriented

$$\frac{1}{2\pi} \Delta_C^0 \arg f(s) = - \frac{1}{2\pi} \Delta_{-C}^0 \arg f(s) = P_C - Z_C$$

Contours



Images



Poles and zeros

$$\begin{aligned} Z_{C_1} &= 1, & P_{C_1} &= 2 \\ Z_{C_2} &= 0, & P_{C_2} &= 2 \\ Z_{C_3} &= 0, & P_{C_3} &= 0 \end{aligned}$$

Total argument

$$\begin{aligned} \left(\frac{1}{2\pi}\right) \Delta^0_{C_1} \arg G(s) &= Z_{C_1} - P_{C_1} = -1 \\ \left(\frac{1}{2\pi}\right) \Delta^0_{C_2} \arg G(s) &= P_{C_2} - Z_{C_2} = 2 \\ \left(\frac{1}{2\pi}\right) \Delta^0_{C_3} \arg G(s) &= Z_{C_3} - P_{C_3} = 0 \end{aligned}$$

Application

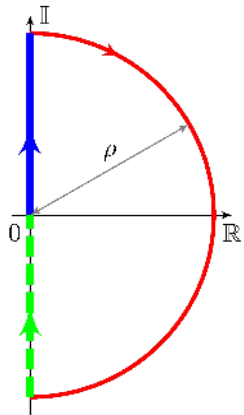
Testing for asymptotic stability

7.5 Stability in the Frequency Domain

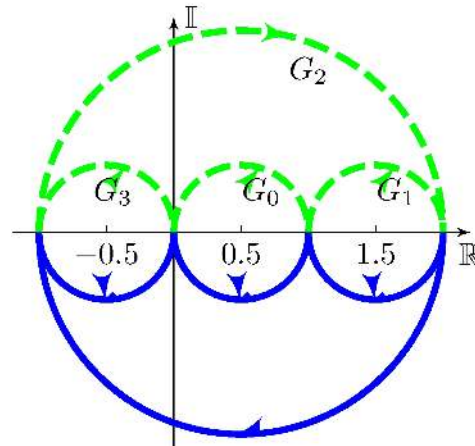
Basic idea

$$\begin{aligned} P_{\Gamma} &= Z_{\Gamma} + \frac{1}{2\pi} \Delta^0_{\Gamma} \arg G(s) \end{aligned}$$

Contour Γ



Polar plot



Transfer-functions

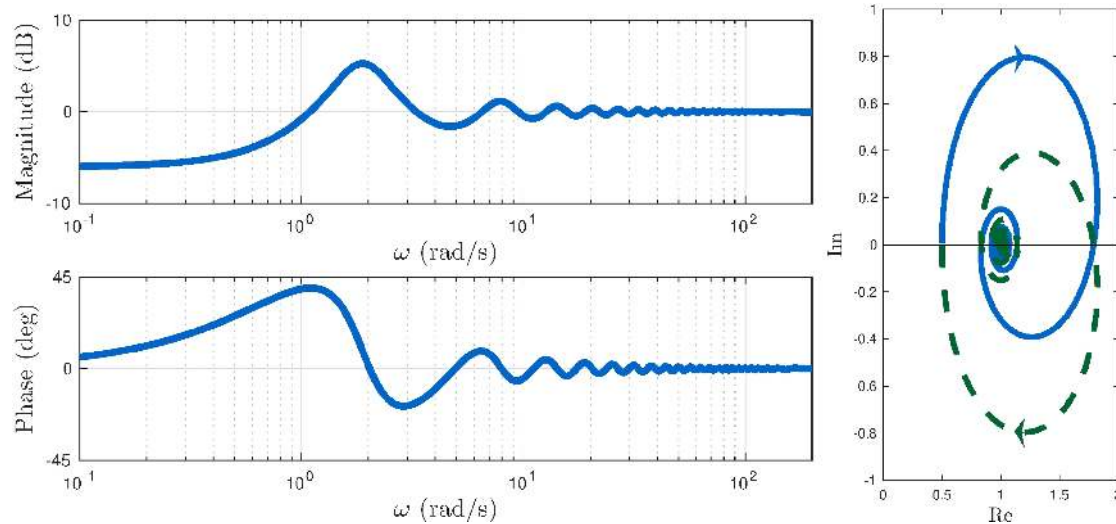
$$\begin{aligned} G_0(s) &= \frac{1}{s+1} \\ G_1(s) &= \frac{s+2}{s+1} = 1 + G_0(s) \\ G_2(s) &= \frac{2-s}{s+1} = G_0(s) - 1 \\ G_3(s) &= \frac{1}{s+1} + 1 = G_0(s) - 1 \end{aligned}$$

$$\begin{aligned} \{Z_{\Gamma}_i &= 0, \quad (2\pi)^{-1} \Delta^0_{\Gamma} \arg G_i(s) = 0, \quad \{P_{\Gamma}_i = 0 + 0 = 0, \quad i = \{0, 1, 3\} \\ \{Z_{\Gamma}_2 &= 1, \quad (2\pi)^{-1} \Delta^0_{\Gamma} \arg G_2(s) = -1, \quad \{P_{\Gamma}_2 = 1 - 1 = 0 \end{aligned}$$

Features

Attention near the origin if $\backslash(G\backslash)$ is strictly proper

Can be used even if $\backslash(G\backslash)$ is not rational, e.g. $\backslash(G(s) = \{(s + 1)\}/\{(s + 1 + e^{-s})\})\backslash)$

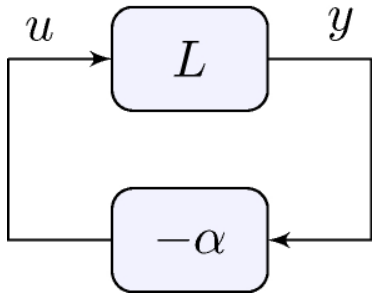


More impressive application

Closed-loop stability analysis: the Nyquist stability criterion

7.6 Nyquist Stability Criterion

Basic idea



Closed loop poles are the zeros of the characteristic equation $1 + \alpha L(s) = 0$, $\alpha > 0$ which are the same as the zeros of $L_\alpha(s) = \frac{1}{\alpha} + L(s)$, $\alpha > 0$ Image of Γ under $(L_\alpha(s))$ is the same as image of Γ under $(L(s))$ shifted by $(\frac{1}{\alpha})$

Example #1

Transfer-function

$$\begin{aligned} L(s) &= \frac{s - 0.5}{s^4 + 2.5 s^3 + 3 s^2 + 2.5 s + 1}, & L_{\alpha}(s) &= \frac{1}{\alpha} + L(s) \end{aligned}$$

Image of Γ under L

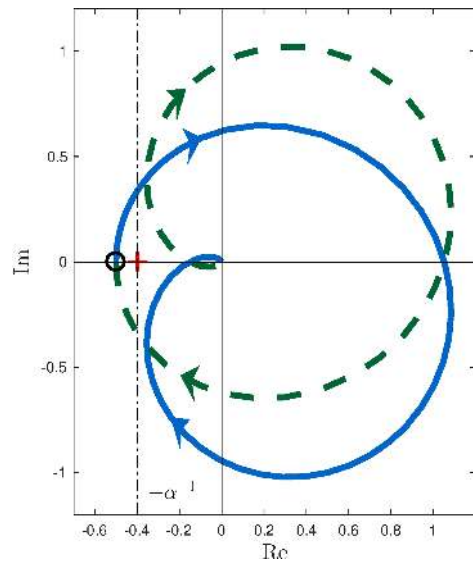
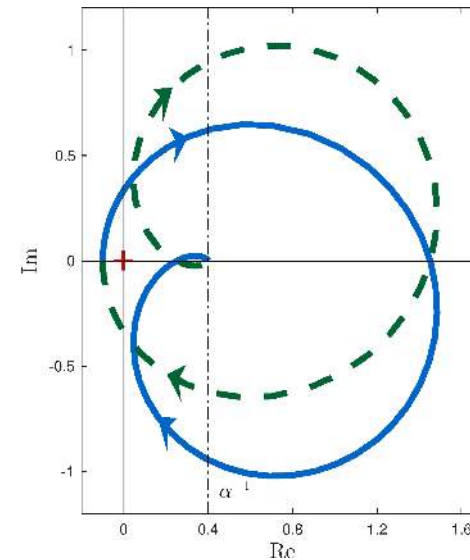


Image of Γ under L_{α} , $(\alpha = 2.5)$



Basic idea

Image of Γ under $(L_\alpha(s))$ is the same as image of Γ under $(L(s))$ shifted by $\frac{1}{\alpha}$

Total argument variation are the same if we shift the origin $\frac{1}{2\pi} \Delta^0_\Gamma$
 $\arg L_\alpha(s) = \frac{1}{2\pi} \Delta^{-1/\alpha}_\Gamma$ $\arg L(s) = P_\Gamma - Z_\Gamma$

Theorem 7.2 (Nyquist)

Closed-loop asymptotic stability if and only if $Z_\Gamma = P_\Gamma - \frac{1}{2\pi} \Delta^{-1/\alpha}_\Gamma$
 $\arg L(s) = 0$

(P_Γ) : # of poles of $(L(s))$ in Γ (unstable open-loop poles)

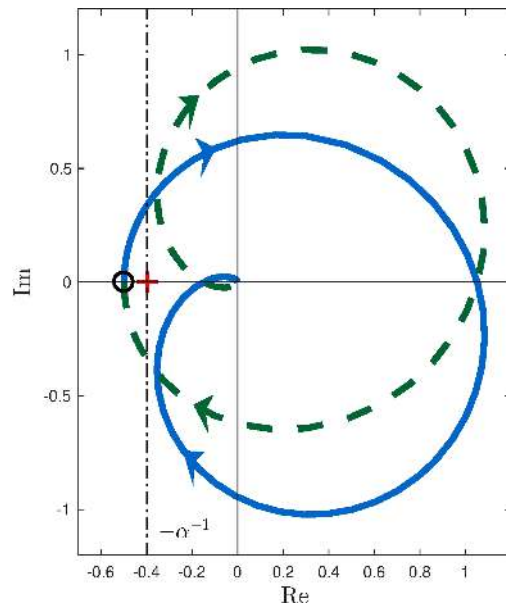
(Z_Γ) : # of poles of $(1 + \alpha L(s))$ in Γ (unstable closed-loop poles)

Example #1

Transfer-function

$$L(s) = \frac{s - 0.5}{s^4 + 2.5s^3 + 3s^2 + 2.5s + 1}$$

Image of Γ
under L



Encirclements of $(-1/\alpha = -0.4)$

$$\frac{1}{2\pi} \Delta_{\Gamma}^{-0.4} \arg L_{\alpha}(s) = -1$$

Closed-loop asymptotic stability? NO

$$Z_{\Gamma} = P_{\Gamma} - \frac{1}{2\pi} \Delta_{\Gamma}^0 \arg L_{\alpha}(s) = 0 - (-1) = 1$$

closed-loop poles in Γ

What if $(-\alpha^{-1} < -0.5)$?

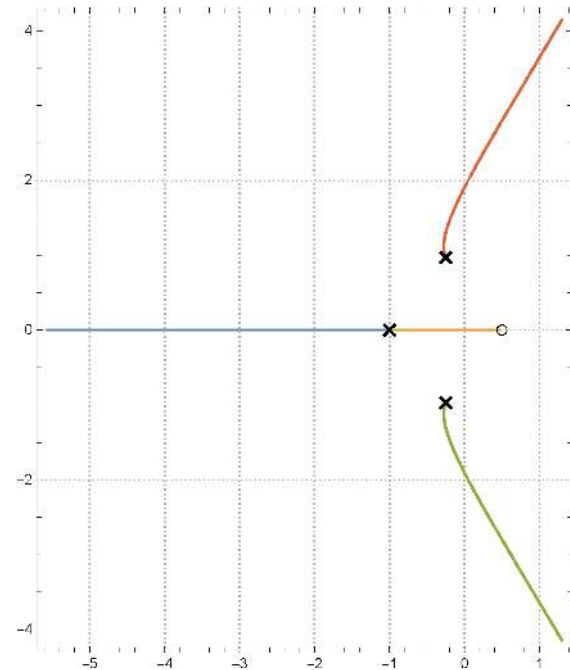
Or if $(-0.1 < -\alpha^{-1} < 0)$?

Example #1

Transfer-function

$$L(s) = \frac{s - 0.5}{s^4 + 2.5s^3 + 3s^2 + 2.5s + 1}$$

Root-locus

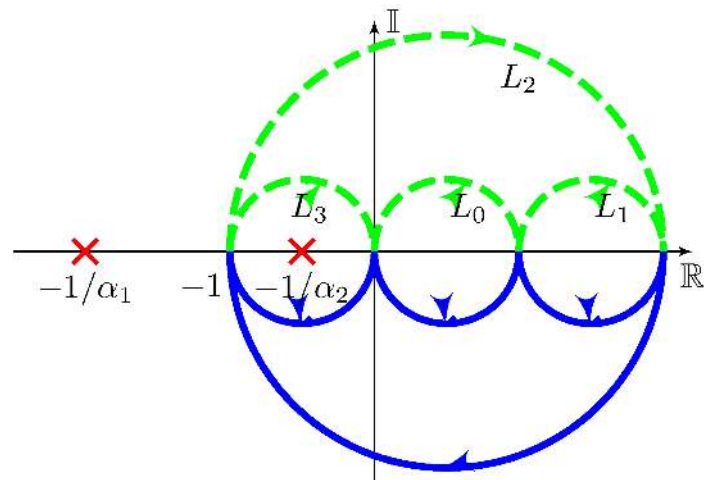


Example #2

Transfer-functions

$$\begin{aligned} L_0(s) &= \frac{1}{s+1}, & L_1(s) &= \frac{s+2}{s+1}, & L_2(s) &= \frac{2-s}{s+1}, & L_3(s) &= \\ & \frac{-s}{s+1} \end{aligned}$$

Nyquist plot



Closed-loop stability

$$(P_{\Gamma} = 0)$$

$$(L_0) \text{ and } (L_1) \quad \frac{1}{2\pi i}$$

$$\Delta^{-1/\alpha}_{\Gamma} \arg L_{\alpha}(s) = 0 \quad \& \quad \implies \quad Z_{\Gamma} = 0$$

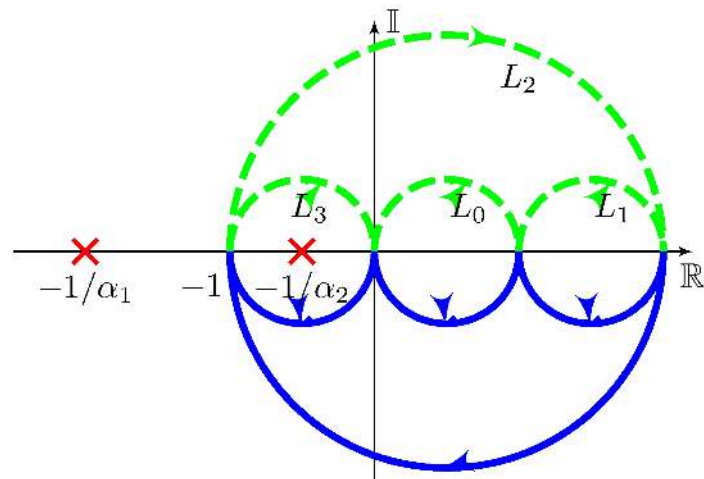
stable for all $(\alpha > 0)$

Example #2

Transfer-functions

$$\begin{aligned} L_0(s) &= \frac{1}{s+1}, & L_1(s) &= \frac{s+2}{s+1}, & L_2(s) &= \frac{2-s}{s+1}, & L_3(s) &= \\ & \frac{-s}{s+1} \end{aligned}$$

Nyquist plot



Closed-loop stability

$$(P_{\Gamma} = 0)$$

(L_0) and (L_1) stable for all $(\alpha > 0)$

(L_2) and (L_3) $\frac{1}{2\pi}$

$\Delta^{-1/\alpha_1}_{\Gamma} \arg L_{\alpha}(s) = 0 \implies$

$Z_{\Gamma} = 0 \implies \frac{1}{2\pi}$

$\Delta^{-1/\alpha_2}_{\Gamma} \arg L_{\alpha}(s) = -1 \implies$

$Z_{\Gamma} = 1$ stable for all $(\alpha = \alpha_1)$

unstable for all $(\alpha = \alpha_2)$

Nyquist Stability Criterion

Theorem 7.2 (Nyquist)

Closed-loop asymptotic stability if and only if
$$Z_{\Gamma} = P_{\Gamma} - \frac{1}{2\pi} \Delta_{\Gamma}^{-1/\alpha} \arg L(s) = 0$$

(P_{Γ}) : # of poles of (G) in (Γ) (unstable open-loop poles)

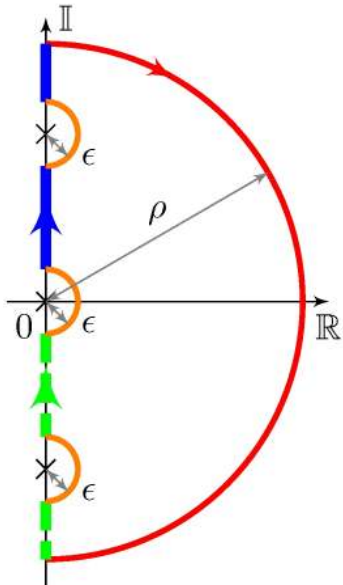
(Z_{Γ}) : # of poles of $(1 + \alpha L(s))$ in (Γ) (unstable closed-loop poles)

Poles on the imaginary axis can be dealt with indentations

Nyquist Stability Criterion

Indented Γ contour

$$L(s) = \frac{F(s)}{(s - j\omega_0)^n}, \quad F(j\omega_0) \neq 0$$



Magnitude

$$|L(j\omega_0 + \epsilon, e^{j\theta})| \approx \epsilon^{-n}, \quad |F(j\omega_0)| \rightarrow \infty, \quad \epsilon \approx 0$$

Phase

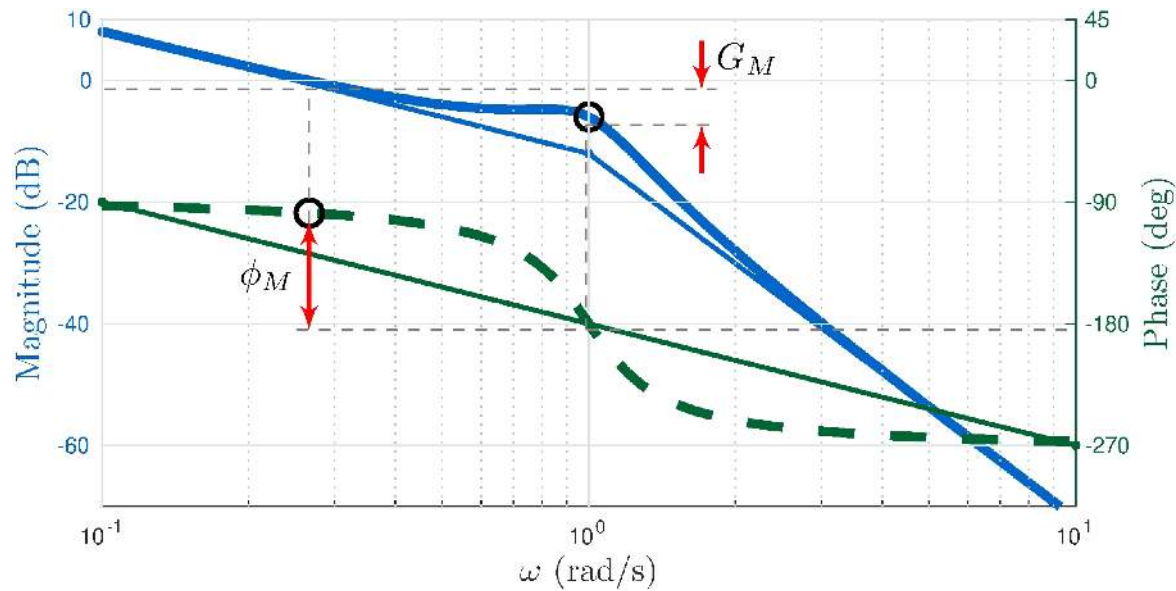
$$\angle L(j\omega_0 + \epsilon, e^{j\theta}) \approx \angle F(j\omega_0) - n\theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad \epsilon \approx 0$$

Example #3

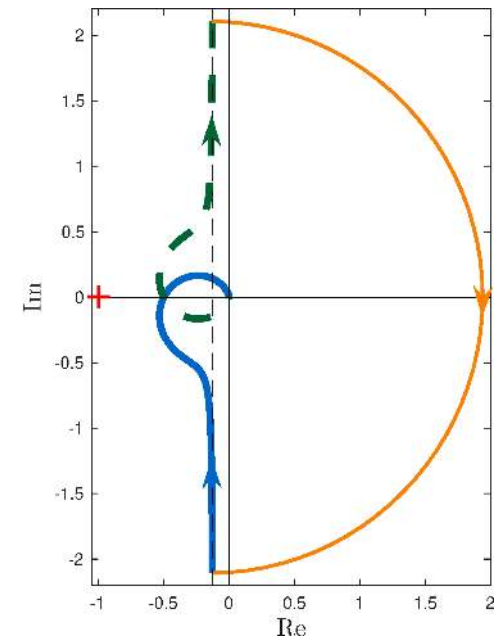
Transfer-function

$$L(s) = \frac{1}{4} \frac{1}{s(s^2 + \frac{s}{2} + 1)}, \quad \text{stable for } \alpha = 1$$

Bode plots



Nyquist plot

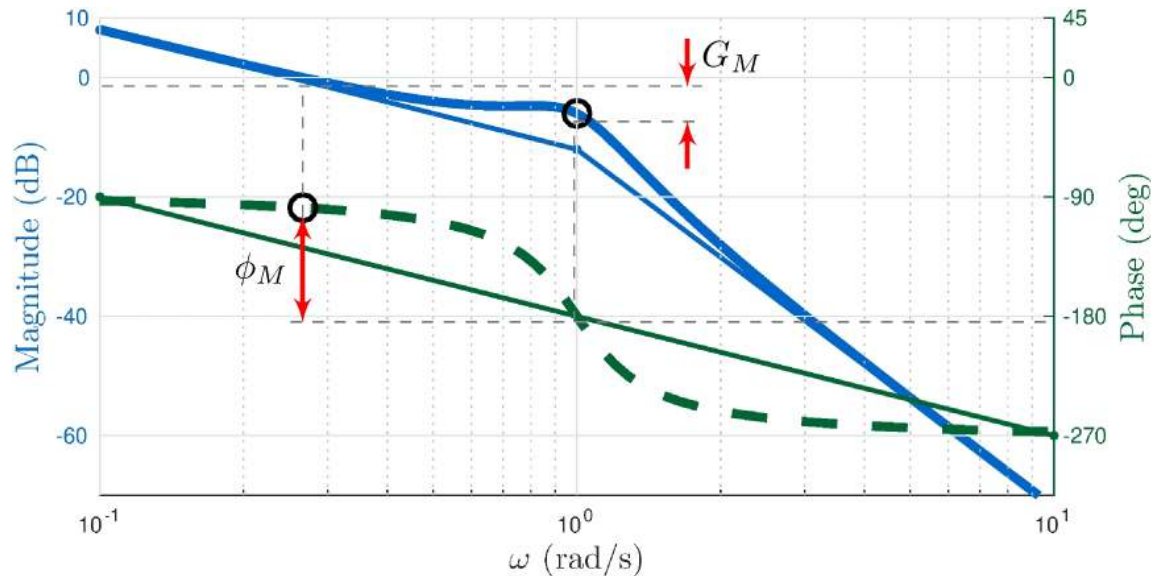


7.7 Stability Margins

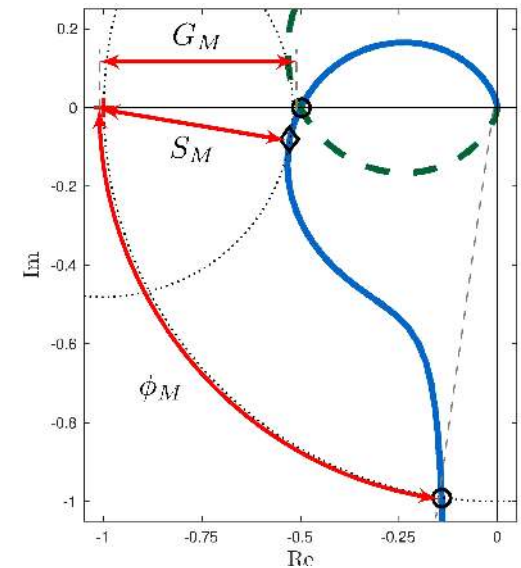
Gain margin

$$G_M = |L(j\omega_G)|^{-1}, \text{ for } \omega_G \text{ s.t. } \angle L(j\omega_G) = \pi$$

Bode plots



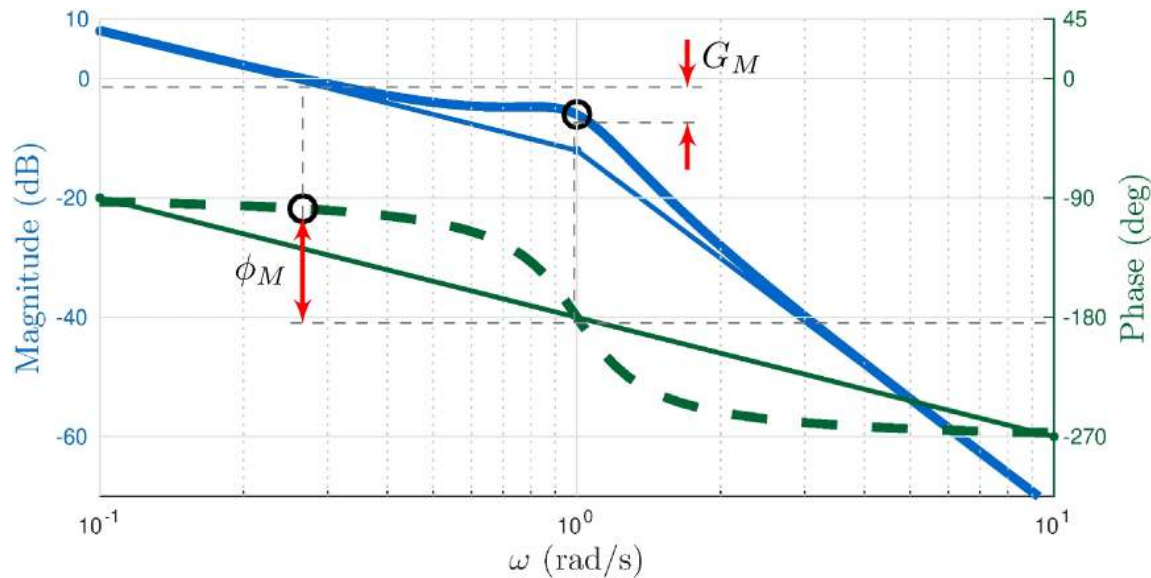
Nyquist plot



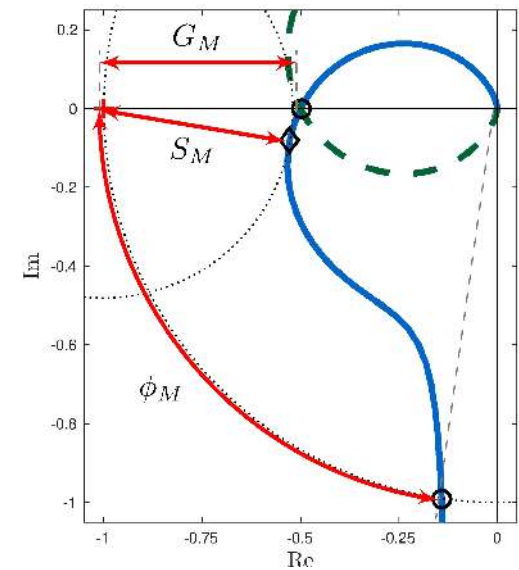
Phase margin

$$\phi_M = \angle L(j\omega_{\phi}) - \pi, \text{ for } |L(j\omega_{\phi})| = 1$$

Bode plots



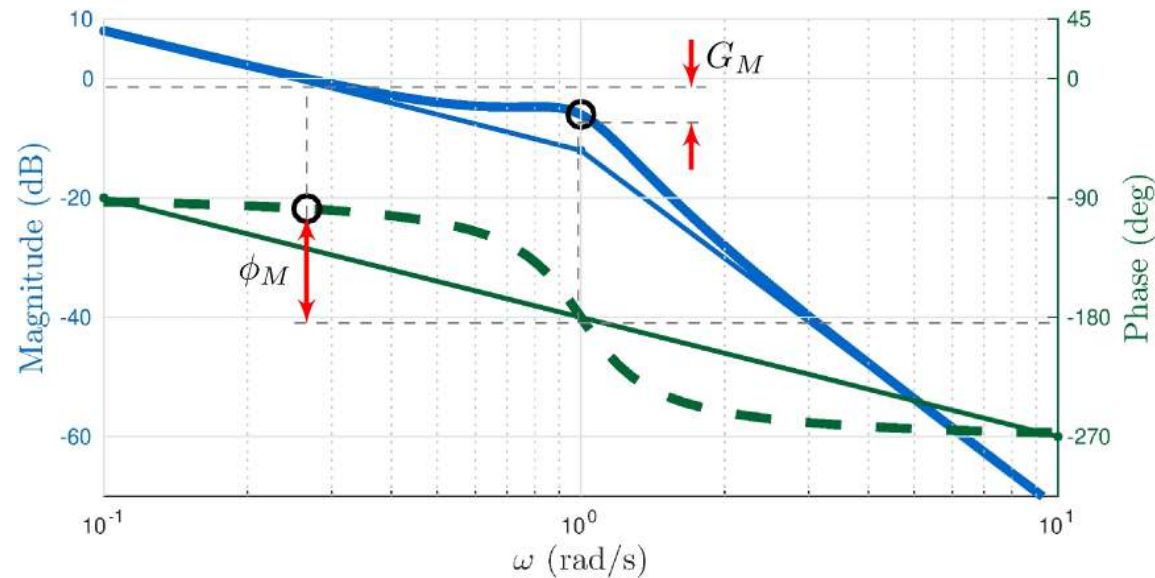
Nyquist plot



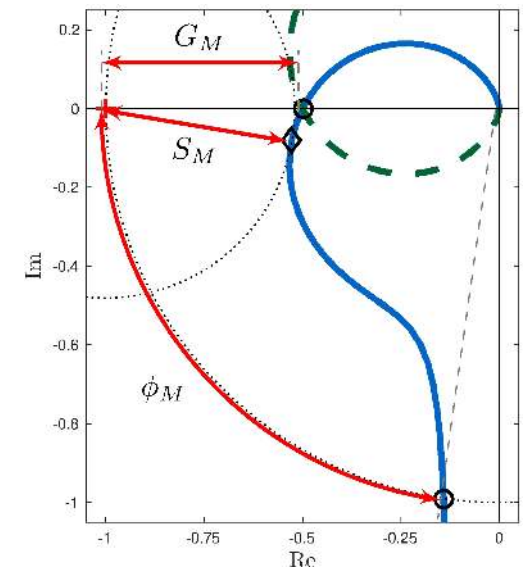
Stability margin

$$\begin{aligned} S_M &= \inf_{\omega} |L(j\omega) - (-1)| = \frac{1}{\sup_{\omega} |S(j\omega)|} = \frac{1}{\|S\|_{\infty}} \end{aligned}$$

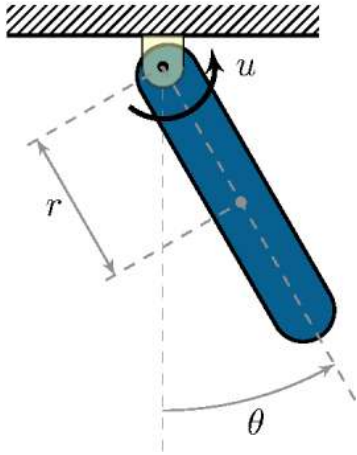
Bode plots



Nyquist plot



7.8 Control of the Simple Pendulum - Part II



Data

```
\[\begin{align*} m &= 0.5 \text{ kg}, & \ell &= 0.3 \text{ m}, & r &= \ell/2 = 0.15 \text{ m}, \\ b &= 0 \text{ kg/s}, & g &= 9.8 \text{ m/s}^2, & J &= 3.75 \times 10^{-3} \text{ kg m}^2, & J_r \\ &= J + m r^2. \end{align*}\]
```

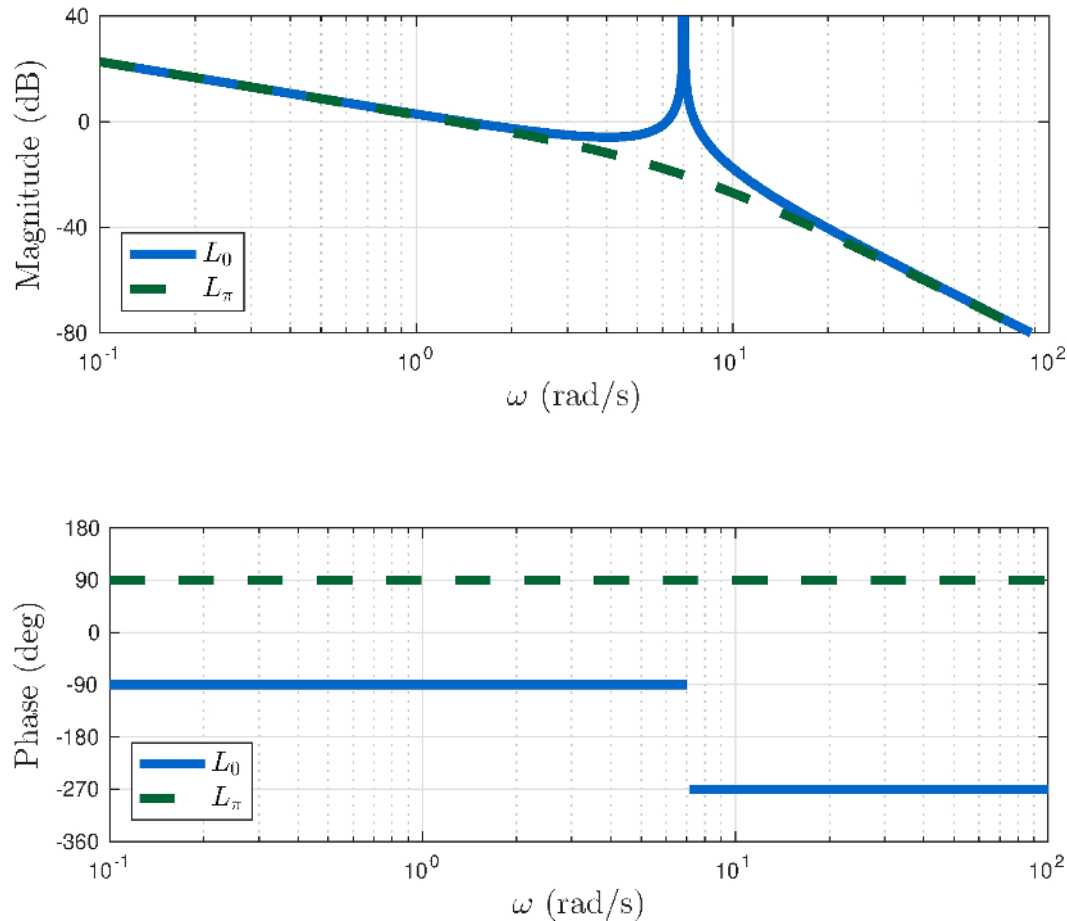
Linearized models

```
\[\begin{align*} G_0(s) &= \frac{1/J_r}{s^2 + (b/J_r)s + m g r/J_r} \approx \frac{66.7}{s^2 + 49}, & G_{\pi}(s) &= \\ &= \frac{1/J_r}{s^2 + (b/J_r)s - m g r/J_r} \approx \frac{66.7}{s^2 - 49} \end{align*}\]
```

Goal

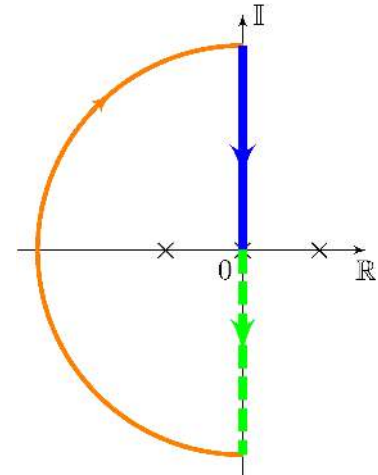
Design a controller with integral action

Control of the Simple Pendulum - Part II

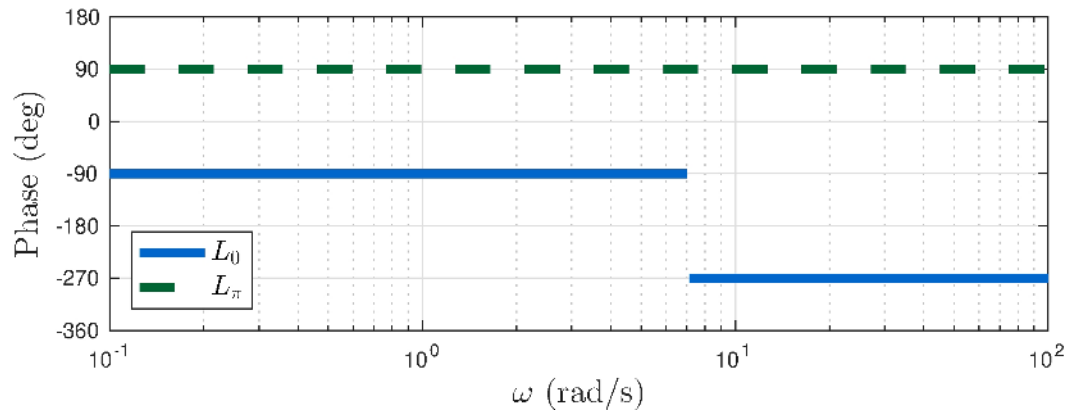
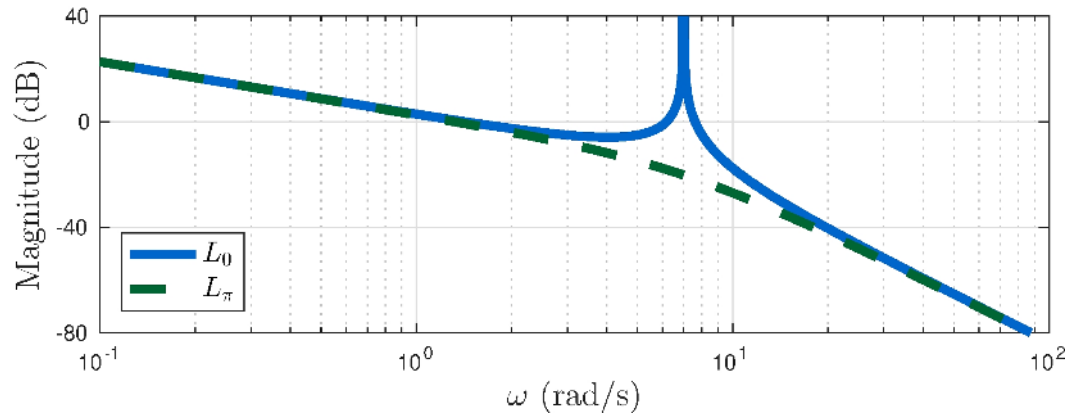


Nyquist analysis

$$\begin{aligned} L_\pi &= \frac{G_\pi(s)}{s} \\ &\approx \frac{-1.36}{s(1 + s/7)(1 - s/7)} \\ Z_\Gamma &= P_\Gamma - \frac{1}{2\pi} \Delta_\Gamma^{-1} \arg L_\pi(s) \\ &= 1 - (-1) = 2 \end{aligned}$$



Control of the Simple Pendulum - Part II



Stability analysis

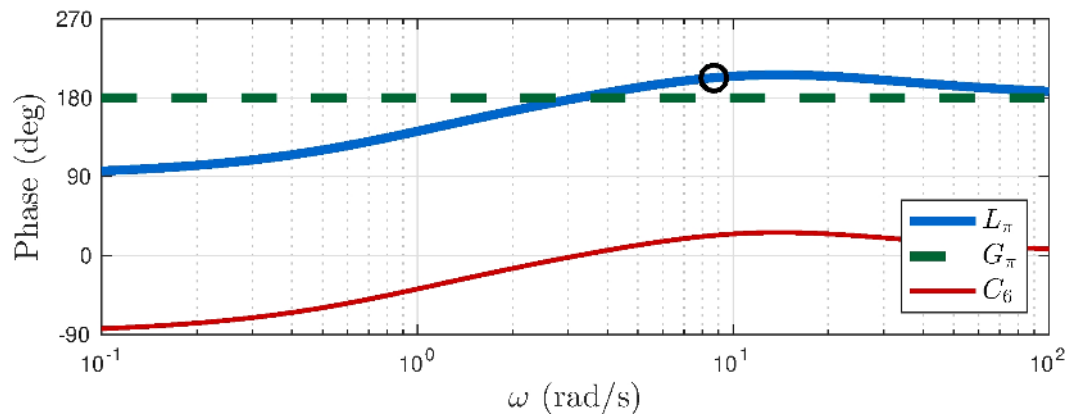
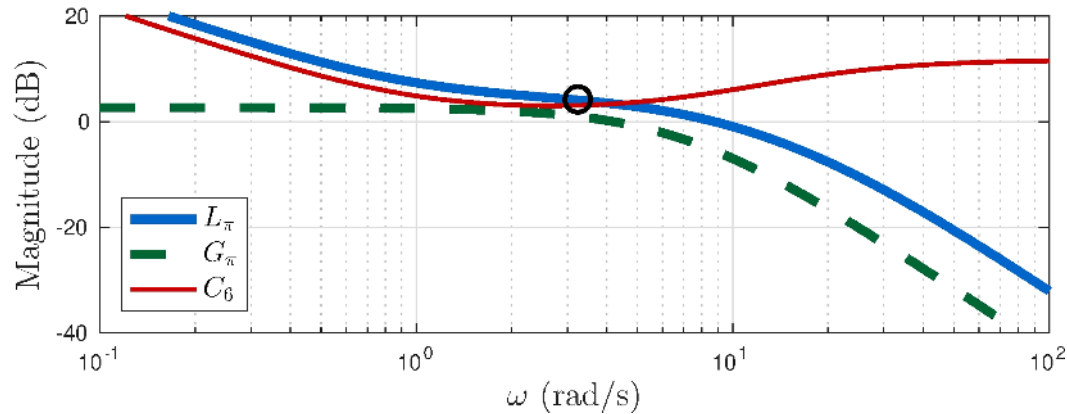
Because $(P_\gamma = 1)$, $(L_\pi(j\omega))$ needs

- one counter-clockwise encirclement
- to cross negative real axis
- magnitude greater than one

Counter-clockwise encirclement requires more than (90°) phase to be added to $(L_\pi(j\omega))$. Controller must have

- at least two stable zeros
- at least one pole for realizability (in addition to the integrator)

Control of the Simple Pendulum - Part II



Controller candidate

$$C(s) = K \frac{(s+z_1)(s+z_2)}{(s+p_1)}$$

with

$$p_1 > z_2 \geq z_1$$

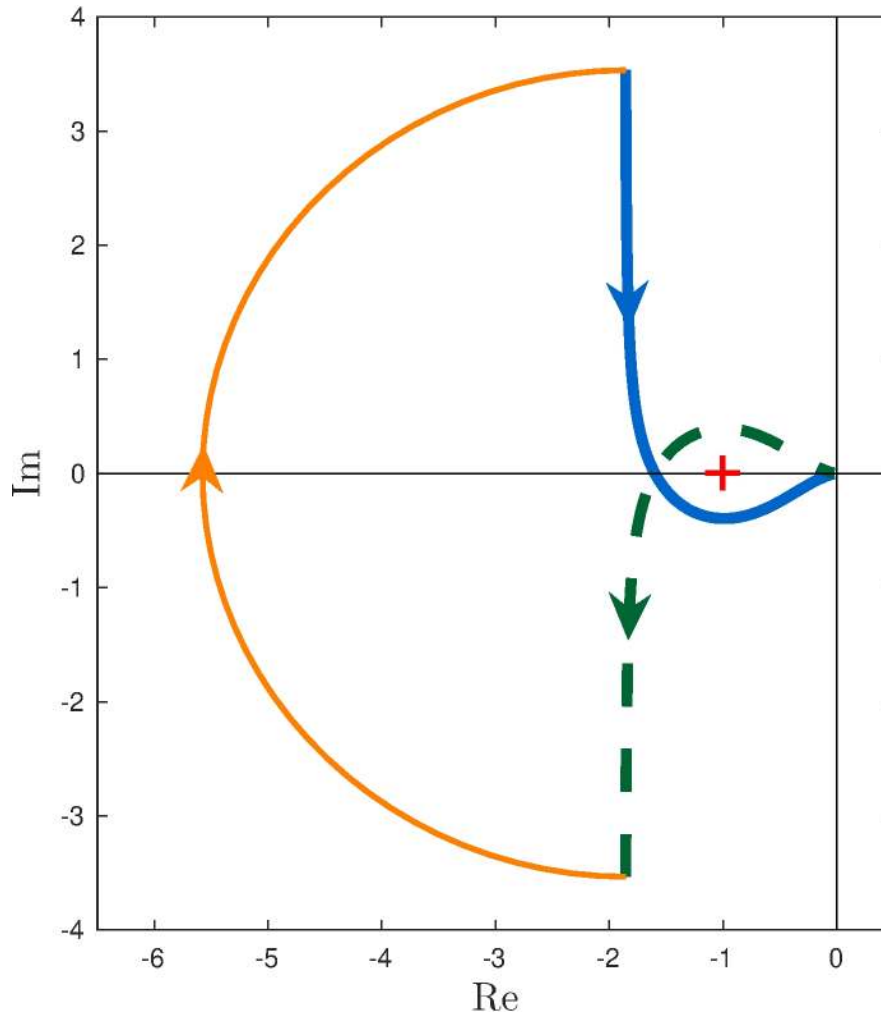
From (C_6)

$$p_1 = 21, K = 3.83, z_2 = 6.86, z_1 = 0.96$$

Margins

$$G_M \approx 0.63 \approx -4 \text{ dB}, \phi_M \approx 23^\circ, S_M = 0.40$$

Control of the Simple Pendulum - Part II



Controller candidate

$$C(s) = K \frac{(s+z_1)}{(s+z_2)(s+p_1)}$$

with

$$p_1 > z_2 \geq z_1$$

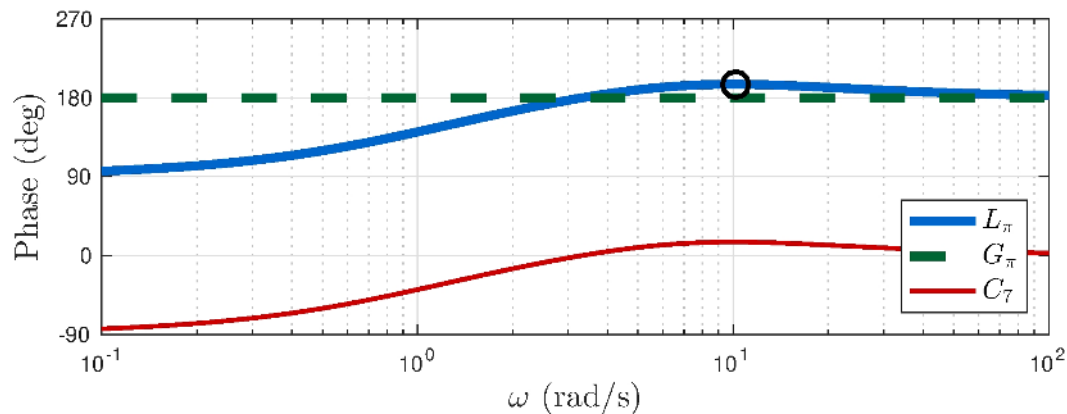
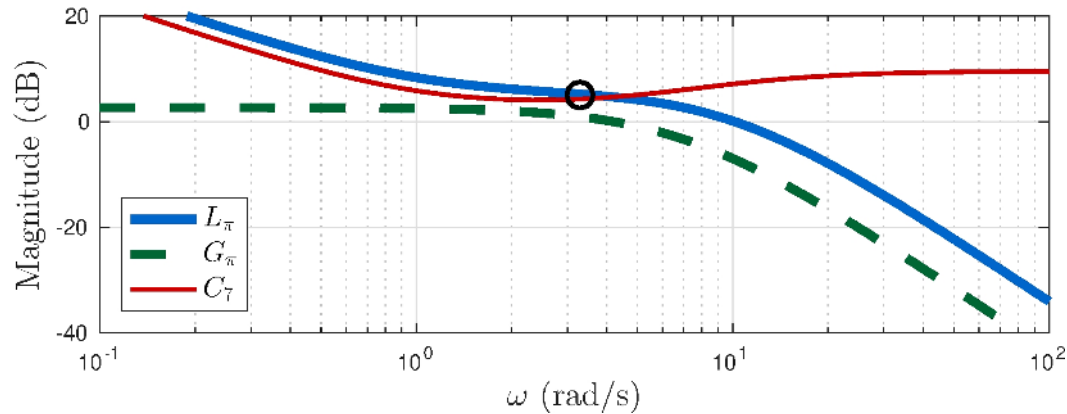
From (C_6)

$$p_1 = 21, K = 3.83, z_2 = 6.86, z_1 = 0.96$$

Margins

$$G_M \approx 0.63 \approx -4 \text{ dB}, \phi_M \approx 23^\circ, S_M = 0.40$$

Control of the Simple Pendulum - Part II



Controller candidate

$$C(s) = K \frac{(s+z_1)(s+z_2)}{(s+p_1)}$$

with

$$p_1 > z_2 \geq z_1$$

From (C_7)

$$p_1 = 11, K = 3, z_2 = 5, z_1 = 1$$

Margins

$$G_M \approx 0.55 \approx -5.2 \text{ dB}, \phi_M \approx 15^\circ, S_M = 0.27$$

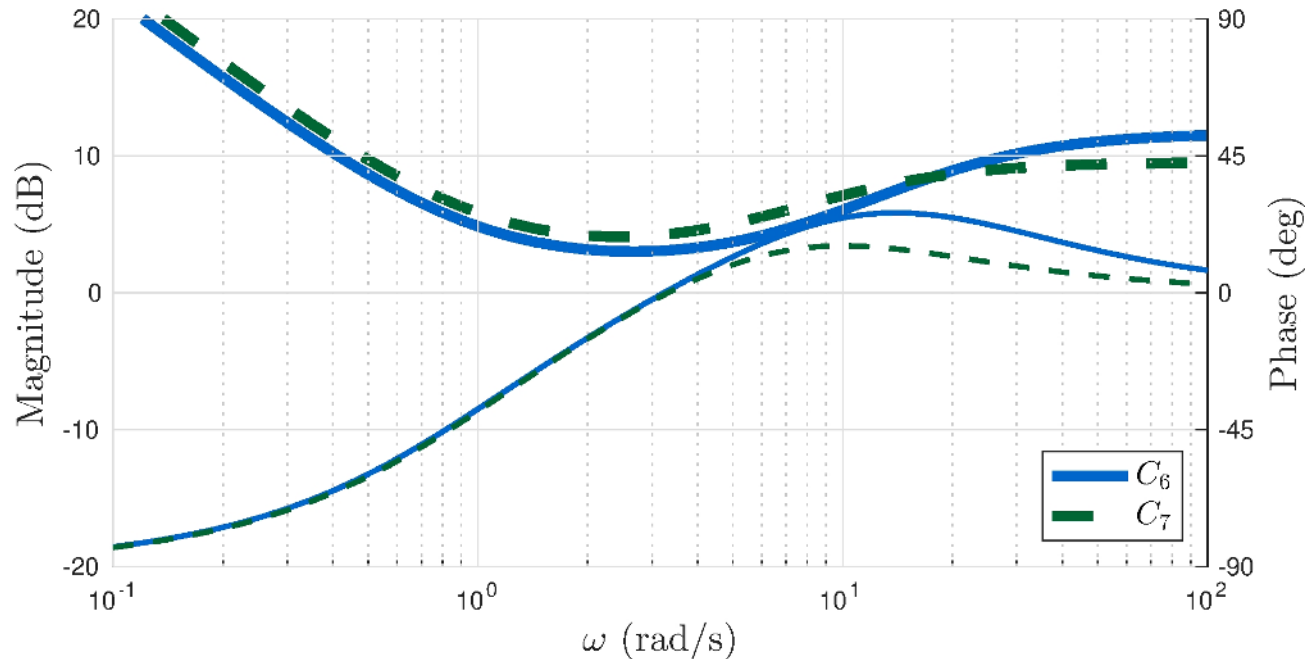
Control of the Simple Pendulum - Part II

Controller (C_6)

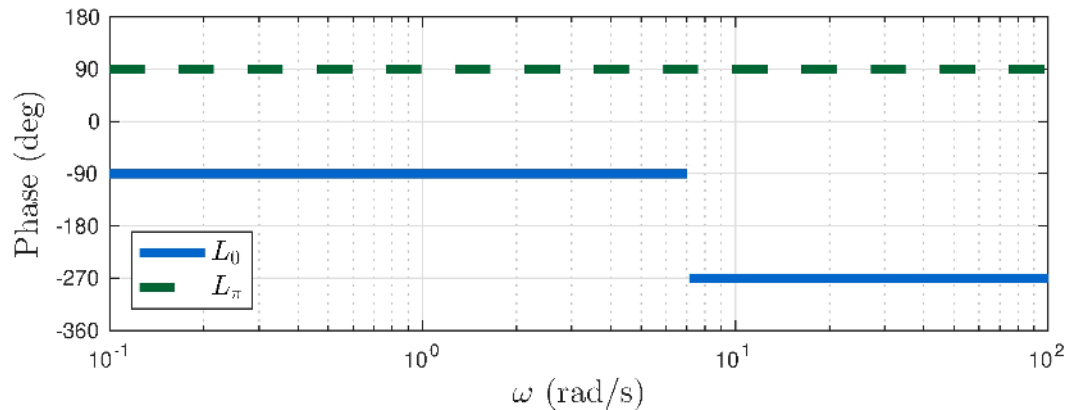
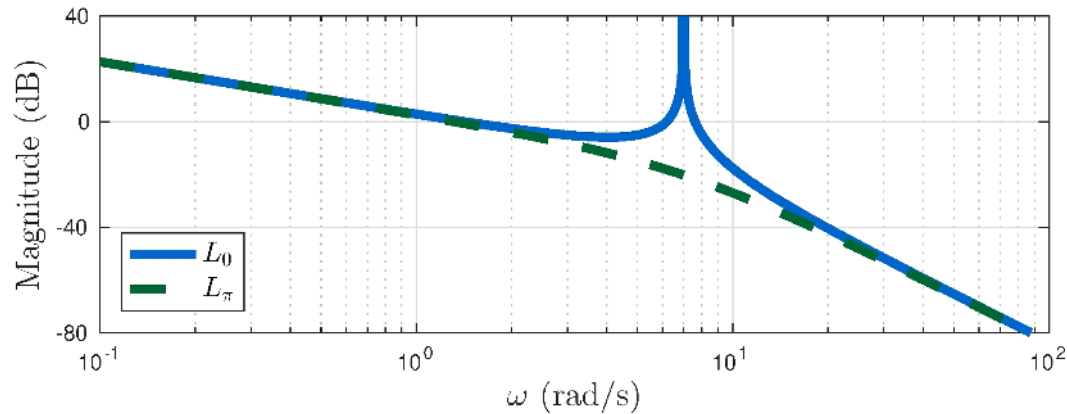
$$C_6(s) = 3.83 \frac{(s+0.96)(s+6.86)}{s(s+21)}$$

Controller (C_7)

$$C_7(s) = 3 \frac{(s+1)(s+5)}{s(s+11)}$$



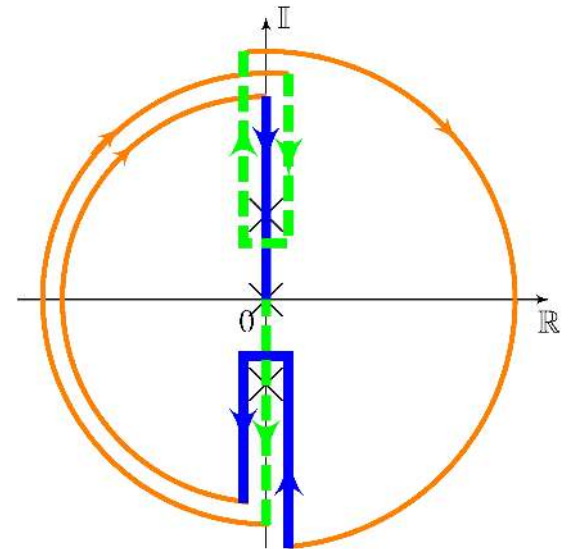
Control of the Simple Pendulum - Part II



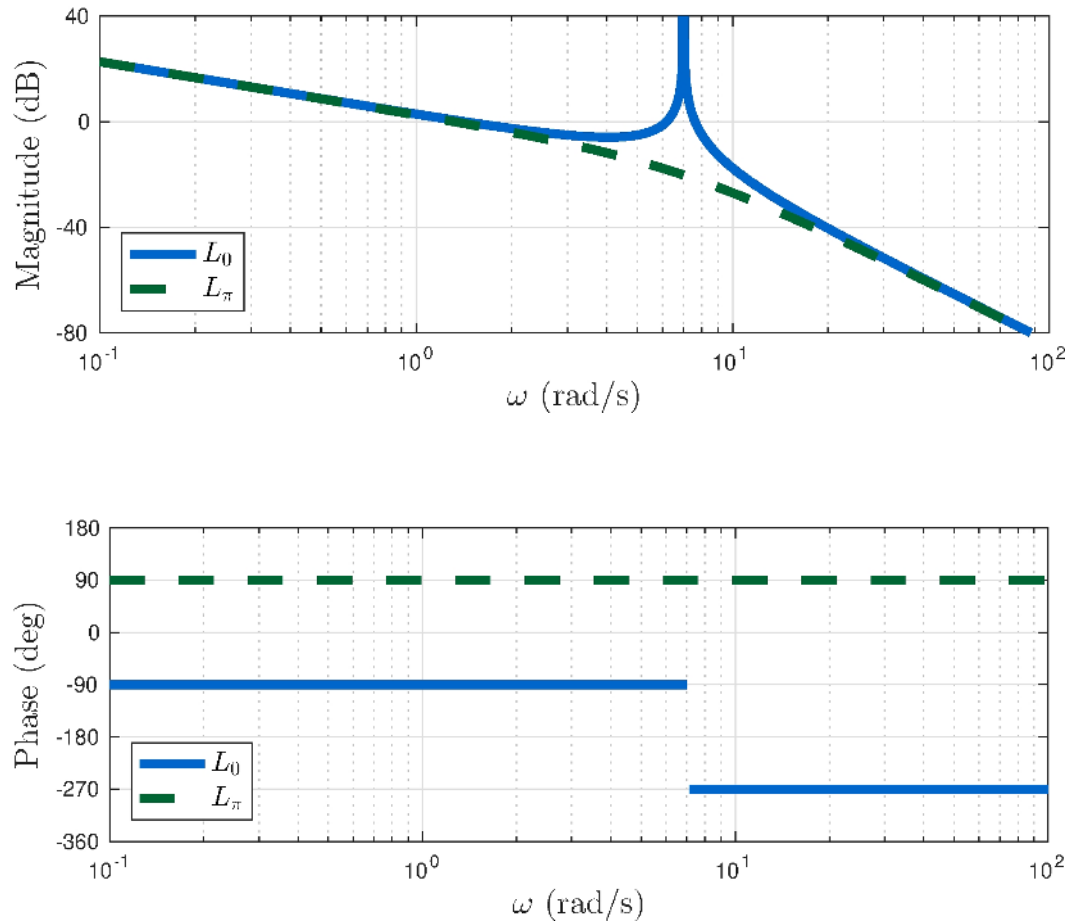
Nyquist analysis

$$L_0 = \frac{G_0}{s} \approx \frac{1.36}{s(1 + j\frac{s}{7})(1 - j\frac{s}{7})}$$

$$Z_\Gamma = P_\Gamma - \frac{1}{2\pi} \Delta_\Gamma^{-1} \arg L_\pi(s) \approx 0 - (-2) = 2$$



Control of the Simple Pendulum - Part II



Stability analysis

Because $\gamma(P_\gamma = 0)$, $(L_0(j\omega))$ needs

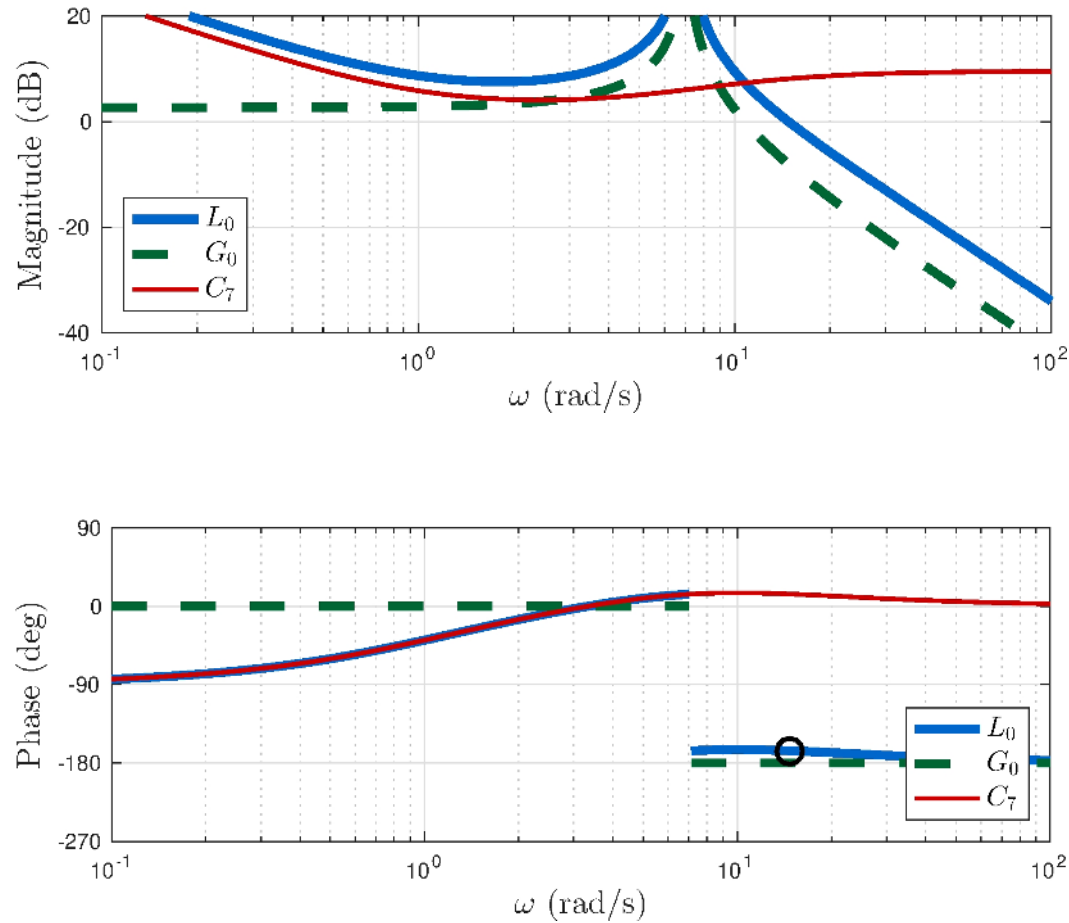
- no encirclements
- to not cross negative real axis

If more than (90°) phase is added to $(L_0(j\omega))$ there will be no crossing of the negative real axis nor encirclements. Controller must have

- at least two stable zeros
- at least one pole for realizability (in addition to the integrator)

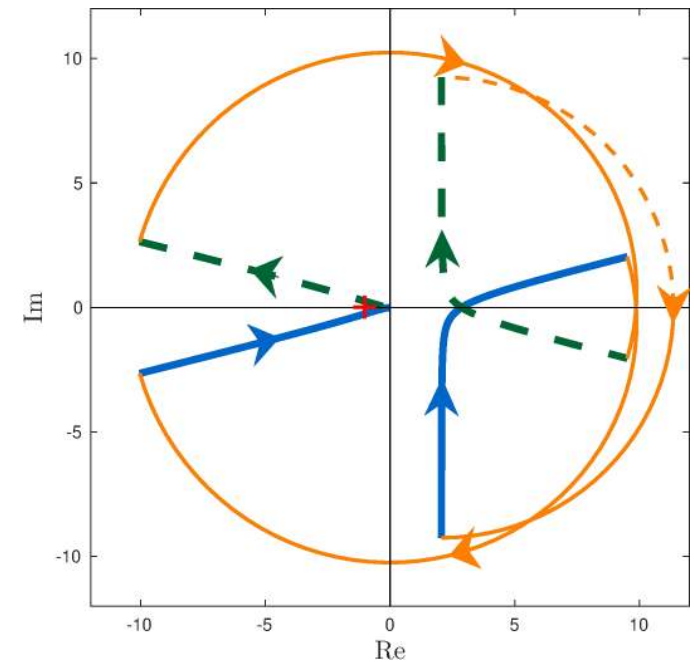
Same specs as for (L_π) !

Control of the Simple Pendulum - Part II



Controller (C_7)

$$C_7(s) = 3 \frac{(s+1)(s+5)}{s(s+11)}$$



Learn more

- [Book website](#)

